

Indefinite Integrals and the Substitution Method

(I) Indefinite Integral: The collection of all

antiderivatives of f is called indefinite integral of f w.r.t 'x'.
(Pg 236).

• It is represented by $\int f(x) dx = F(x) + c$
where c is any arbitrary constant.

• The connection between antiderivatives and the definite integral stated in the Fundamental Theorem now explains this notation:

$$\begin{aligned} \int_a^b f(x) dx &= F(b) - F(a) \quad (\because 2^{nd} \text{ Fundamental Theorem}) \\ &= [F(b) + c] - [F(a) + c] \\ &= [F(x) + c]_a^b \quad (\text{Example 1}) \\ &= \left[\int f(x) dx \right]_a^b \quad \text{Topic 5.6 (Pg 297)} \end{aligned}$$

- When finding the indefinite integral of a function 'f', remember that it always includes an arbitrary constant 'c'.
- A definite integral $\int_a^b f(x) dx$ is a number.
- An indefinite integral $\int f(x) dx$ is a function plus an arbitrary constant 'c'.